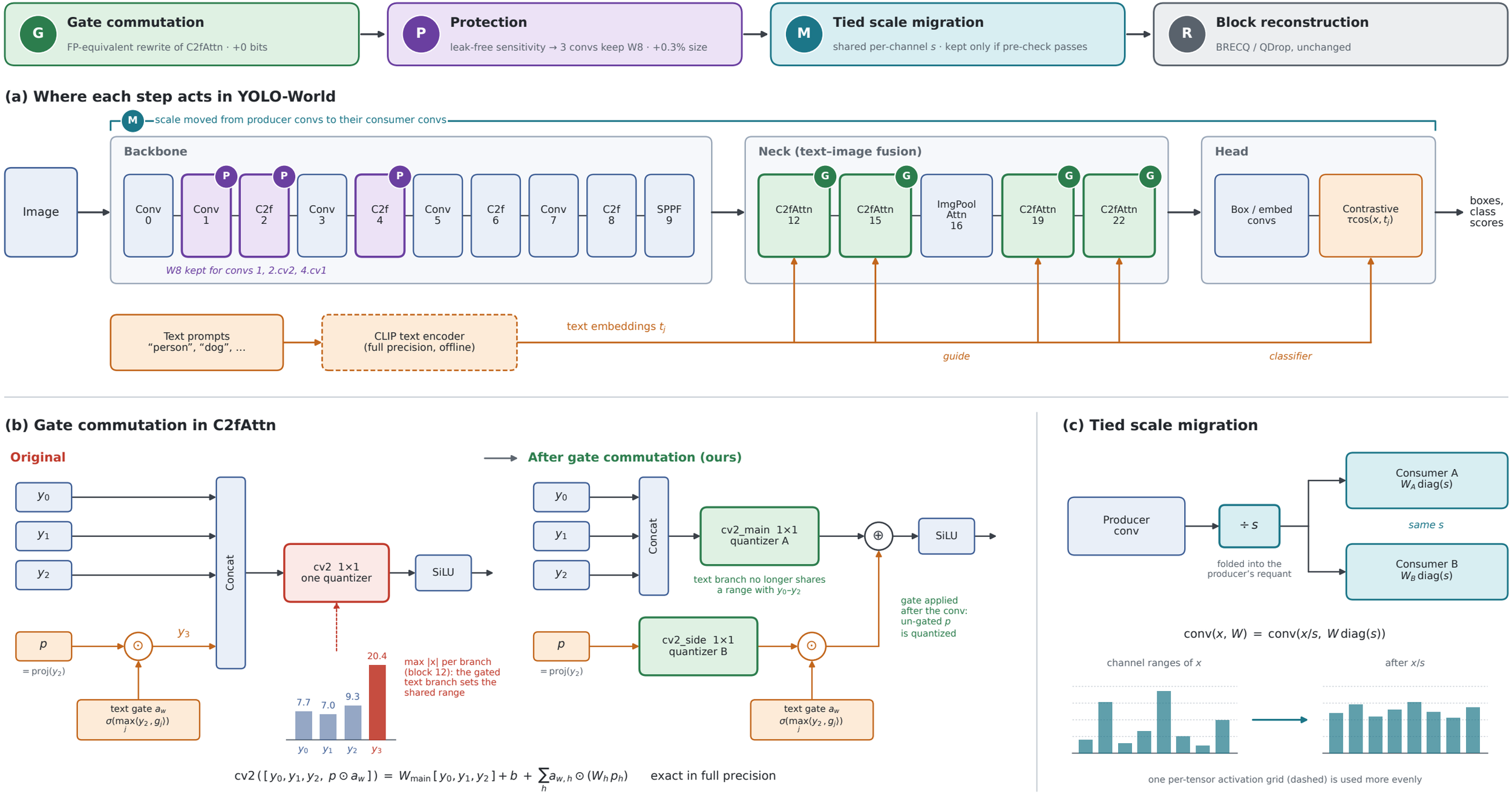
G
Gate commutation
FP-equivalent rewrite of C2fAttn · +0 bits
P
Protection
leak-free sensitivity → 3 convs keep W8 · +0.3% size
M
Tied scale migration
shared per-channel s · kept only if pre-check passes
R
Block reconstruction
BRECQ / QDrop, unchanged
(a) Where each step acts in YOLO-World
M
scale moved from producer convs to their consumer convs
Backbone
Image
Conv 0
Conv 1
C2f 2
Conv 3
C2f 4
Conv 5
C2f 6
Conv 7
C2f 8
SPPF 9
W8 kept for convs 1, 2.cv2, 4.cv1
Neck (text-image fusion)
C2fAttn 12
C2fAttn 15
ImgPool Attn 16
C2fAttn 19
C2fAttn 22
Head
Box / embed convs
Contrastive τcos(x, t_j)
boxes, class scores
Text prompts "person", "dog", ...
CLIP text encoder (full precision, offline)
text embeddings t_j
guide
classifier
(b) Gate commutation in C2fAttn
Original
After gate commutation (ours)
y0
y1
y2
Concat
cv2 1×1 one quantizer
SiLU
p
= proj(y2)
y3
text gate a_w
σ(max_j(y2 , g_j))
7.7
7.0
9.3
20.4
max |x| per branch (block 12): the gated text branch sets the shared range
cv2_main 1×1 quantizer A
text branch no longer shares a range with y0–y2
cv2_side 1×1 quantizer B
gate applied after the conv: un-gated p is quantized
cv2([y0, y1, y2, p ⊙ a_w]) = W_main[y0, y1, y2] + b + Σ_h a_w,h ⊙ (W_h p_h)
exact in full precision
(c) Tied scale migration
Producer conv
÷ s
folded into the producer's requant
Consumer A W_A diag(s)
same s
Consumer B W_B diag(s)
conv(x, W) = conv(x/s, W diag(s))
channel ranges of x
after x/s
one per-tensor activation grid (dashed) is used more evenly